[INCH-POUND]
A-A-2937
November 27, 1995
SUPERSEDING
GGG-R-180
January 21, 1992

COMMERCIAL ITEM DESCRIPTION

REAMER, HAND, ADJUSTABLE

The General Services Administration has authorized the use of this Commercial Item Description for procurement of the Type III, Class 1 reamers specified in GGG-R-180 for all federal agencies.

1. SCOPE

This Commercial Item Description covers adjustable hand reamers used in reaming hole diameters to specified tolerances.

2. CLASSIFICATION

Not applicable.

3. SALIENT CHARACTERISTICS

3.1 Design. The adjustable hand reamer shall be of the straight-fluted inserted blade type.

3.2 Material. The body and adjusting nut part of the reamer shall be made of carbon steel and the cutting blades shall be made of high speed steel.

3.3 Construction. The reamer shall be fitted with removable cutting blades, which shall be adjustable for reaming holes of any size within the range for which the reamer is designated. The reamer shall have straight, round shanks with square ends. The shank shall be concentric with the cutting edges within 0.001 inch total indicator reading (tir) as measured between centers. The body portion of the reamer shall contain slots for holding the blades. The reamer shall be constructed to permit an adjustment of not less than 0.035 inch between the minimum and maximum size attainable with any one set of blades. The reamer assembly shall be capable of reaming round, straight, and smooth holes free of chatter marks within any size range. The pilot bushing of the reamer shall have a surface roughness of not more than 25 micro-inches, arithmetical average (aa). The reamer shall be similar to figure 1 and conform to table 1.

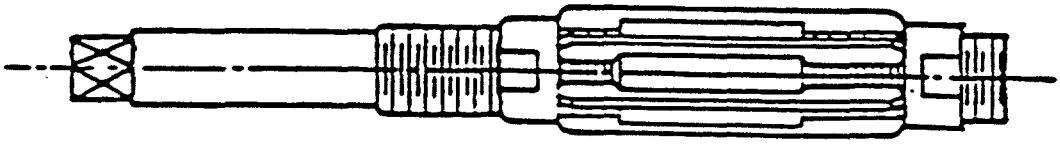


Figure 1. Adjustable, straight-fluted inserted blade reamer

Beneficial comments, recommendations, additions, deletions, clarification, etc. and any data which may improve this document should be sent to: General Services Administration, Federal Supply Service, Tools and Appliances Commodity Center (6FETE-CO), Washington, DC 20406

FSC 5110

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Nominal Size	Range of Expansion		Length of Blades 1/		Length Overall		Number of Blades	Thread Sizes	NSN 5110-00-
Size Letters	Min. In.	Max. In.	Min. In.	Max. In.	Min. In.	Max. In.	Min.		
8/A	1/4	9/32	1 11/32	1 5/8	3 1/4	3 1/2	4	Optional	
7/A	9/32	5/16	1 11/32	1 5/8	3 1/2	3 3/4	4	Optional	
6/A	5/16	11/32	1 1/2	1 3/4	4	4 1/8	4	Optional	
5/A	11/32	3/8	1 1/2	1 3/4	4 1/4	4 3/8	4	Optional	
4/A	3/8	13/32	1 1/2	1 7/8	4 1/2	4 3/4	4	Optional	
3/A	13/32	7/16	1 1/2	1 7/8	4 3/4	5	6	Optional	
2/A	7/16	15/32	1 5/8	1 7/8	5	5 1/4	6	Optional	
A	15/32	17/32	1 1/2	2	4	5 1/2	6	Optional	260-1705
B	17/32	19/32	1 5/8	2 1/8	4 1/2	6	6	Optional	260-1706
C	19/32	21/32	2	2 1/2	5	6 1/2	6	Optional	260-1707
D	21/32	23/32	2	2 3/4	5 1/2	7	6	Optional	260-1708
E	23/32	25/32	2 1/4	3	6	7 3/4	6	Optional	289-1011
F	25/32	27/32	2 1/4	3 1/2	6 1/2	8 1/2	6	Optional	260-1717
G	27/32	15/16	2 3/4	3 1/2	7	9	6	Optional	260-1728
H	15/16	1 1/16	2 3/4	3 1/2	7 1/4	9 3/4	6	Optional	260-1727
I	1 1/16	1 3/16	3	3 7/8	7 3/4	10 3/4	6	Optional	
J	1 3/16	1 11/32	3 1/8	4	8	11 3/4	6	Optional	
K 2/	1 11/32	1 17/32	3 3/8	4 1/2	9	12 3/4	6	Optional	
	1 11/32	1 1/2	3 3/8	4 1/2	9	12 3/4	6	Optional	
L 2/	1 17/32	1 25/32	3 1/2	4 1/2	13	14	6	Optional	
	1 1/2	1 13/32	3 1/2	4 1/2	13	14	6	Optional	
M 2/	1 25/32	2 3/32	4	5	15	16	6	Optional	
	1 13/16	2 7/32	4	5	15	16	6	Optional	
N 2/	2 3/32	2 15/32	4 13/16	5 3/4	17	18	6	Optional	
	2 7/32	2 3/4	4 13/16	5 3/4	17	18	6	Optional	
O 2/	2 15/32	2 31/32	5 3/16	6 1/2	19	20	6	Optional	
	2 3/4	3 11/32	5 3/16	6 1/2	19	20	6	Optional	

Table 1

1/ Length of blades to be measured between adjusting nuts with the reamer at maximum size.
2/ Either range is acceptable . However, when provided in sets, the ranges furnished shall be such that complete dimensional coverage is provided.

3.4 Blades. The blades shall be accurately machined and ground in sets to properly fit slots provided on the body of the reamer. Tolerances for the length of individual sets of blades shall not exceed plus or minus 0.0025 inch. The width of the blades shall be uniform and interchangeable for the same size and make of reamer. Cutting surfaces of blades shall be concentric with the axis of the body in all positions throughout the range of adjustment. The adjustment of the blades shall be facilitated by an adjusting nut located at opposite ends of the blades. Blade ends shall be ground and rounded so that the adjusting nut, or washer when used, is not damaged by the blade ends. When required to be measured, the length of the blade shall be measured between adjusting nuts.

3.5 Adjusting nuts. Adjusting nuts shall have at least two flat sides, or other suitable means not requiring special tools or wrenches, for the expansion adjustment.

3.6 Hardness. The full length of the cutting edges of the reamer shall be hardened and tempered to a Rockwell hardness value from C61 to C67. When required to be tested, the reamer shall be tested for hardness in accordance with ASTM E18.

3.7 Threads. The reamer shall have fine threads in accordance with FED-STD-H28.

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3.8 Interchangeability. All parts shall be designed and manufactured to definite standards, clearances, and tolerances in order that any such parts of particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit.

3.9 Performance. The reamer shall cut freely and produce smooth and accurate holes when subjected to hand reaming ten holes in 1/2 inch thick steel conforming to FED-STD-66, AISI Number 4140 or 4150, and having a Brinell hardness value from 235 to 285. The holes shall be prepared 0.002 to 0.004 inch smaller than the size hole desired.

3.10 Workmanship. Details of workmanship shall be in accordance with the best commercial standards and practices. Paints, coatings, platings, and finishes shall be smooth adherent, continuous, and not stained or discolored. External surfaces shall be free of tool and gouge marks, nicks, or other surface imperfections. The item shall be free from manufacturing workmanship defects (e.g. sharp or rough external edges, corners or surfaces) and material workmanship defects (e.g., pits, rips, fins, burrs, tears nodules, cracks, blisters) which may adversely impact the item's serviceability, durability, safety or appearance.

3.11 Identification marking. The reamer shall be marked with the manufacture's name or identifying symbol, and the state or country of manufacture, unless otherwise specified. All identification markings shall be engraved, etched, molded, or indented directly on the item's surface in such a manner that it remains clearly legible throughout the life of the item. The reamer blades shall be marked in sets with the reamer's nominal size. In addition, the reamer shall be marked with the words "High Speed" or the abbreviation "HS".

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the tolerances specified (using conversion tables contained in the latest revision of Federal Standard 376) and all other requirements of this document are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch-pound units, a request should be made to the Contracting Officer to determine if the product is acceptable.

5. QUALITY ASSURANCE PROVISIONS

5.1 Contractor certification. The contractor shall provide certification and maintain substantiating evidence that all items being offered for acceptance meet the requirements of this Commercial Item Description and conform to the manufacturer's drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as provided for under the provisions of the contract.

5.2 Responsibility for inspection. Unless otherwise specified, the contractor is responsible for the performance of all inspection requirements and may use any commercial facilities (including the contractor's own facilities) suitable for performance of the inspection requirements, unless disapproved by the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure the item conforms to the specified requirements.

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5.3 Bid sample(s). When a bid sample requirement is specified in the solicitation, the bid sample(s) shall be inspected for all salient characteristics. A failure of any bid sample to meet the salient characteristics shall be cause for rejection.

5.4 First article sample(s). When a first article requirement is specified in the contract, the first article sample(s) shall be inspected for all salient characteristics. A failure of any first article sample to meet the salient characteristics shall be cause for rejection.

6. PACKAGING

Preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

(This section contains information of a general or explanatory nature that is not mandatory.)

7.1 Addresses for obtaining copies of reference documents.

Federal Specifications: Federal Supply Service Bureau, Specification Section, Suite 8100, 470 L'Enfant Plaza, SW, Washington, DC, 20407.

Federal Acquisition Regulations (FAR): Government Printing Office Superintendent of Documents, Washington, DC 20401-9371.

ANSI/ASQC Z1.4: American Society for Quality Control, PO Box 3005, 611 E. Wisconsin Ave., Milwaukee, WI 43201-4606.

ASTM E18.: American Society for Testing and Materials, 1916 race Sty., Philadelphia, PA 19103.

7.2 Ordering data. Purchasers should select the preferred options permitted herein and shall include the following information in procurement documents:

(a) Title, number, and date of this Commercial Item Description.
(b) Nominal size of reamer.
(c) When applicable, the appropriate NSN.
(d) If bid samples are required and the number of samples required.
(e) If first article samples are required and the number of samples required.
(f) If lot sampling inspections are required and the applicable lot sampling requirements. For example:

Sampling for inspection. Sampling for inspection (both examinations and testing) shall be in accordance with ANSI/ASQC Z1.4.

Examinations. Each sample tool shall be examined for all salient characteristics. The Inspection Level shall be S-3 with an Acceptable quality level (AQL) of 2.5 percent defective.

Testing. Each sample tool shall be tested for all salient characteristics. The Inspection Level for the hardness test shall be S-2 with an Acceptable quality level (AQL) of 2.5 percent defective. The Inspection Level for the performance test shall be S-2 with an Acceptable quality level (AQL) of 4.0 percent defective.

(h) What preservation, packaging, and marking are required.

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7.3 National Stock Numbers (NSNs). Applicable NSNs are as shown in Table 1. NSNs shown may not be indicative of all possible NSNs associated with this CID.

Military Interests:
None: DoD has no registered interest in revisions and notices to this Commercial Item Description until further notice.

Preparing Activity
GSA - FSS

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